Supplementary Material

Dataset Description: Germany COVID-19 Surveillance Data

Language: Python
Data: Germany COVID-19 Surveillance Data
Date: March 2026

S1. Dataset Overview

The Germany dataset used in this study comprises weekly aggregated counts of two epidemiological indicators for COVID-19 in Germany, covering **Week 45 of 2021 to Week 18 of 2023** (78 observations in total). The data were extracted from *SurvStat@RKI 2.0* (https://survstat.rki.de), the official epidemiological surveillance platform operated by the Robert Koch Institute (RKI) of Germany, which compiles nationally reported case data in accordance with the German Infection Protection Act (*Infektionsschutzgesetz*, IfSG). A detailed epidemiological description of the German pandemic data and the reporting framework can be found in Chukhrova et al. (2024).

Note: The CSV file provided covers the Phase II monitoring period (Week 45/2021 – Week 18/2023). An additional Phase I training set (Week 10/2020 – Week 44/2021, approximately 88 weeks) was used for SPC control chart baseline construction; this segment is not included in the provided file as it corresponds to the historical in-control period described in the paper (Section 5.2.1).

Property	Value
Data source	SurvStat@RKI 2.0 (Robert Koch Institute, Germany)
Geographic scope	Germany (national aggregate)
Temporal resolution	Weekly (ISO calendar weeks)
Coverage period (provided file)	Week 45 of 2021 – Week 18 of 2023
Total observations (rows)	78 weeks
Number of features (columns)	2 (excluding week index)
Missing values	None (complete records across all features and weeks)
Phase I training period (not in file)	Week 10 of 2020 – Week 44 of 2021 (~88 weeks)

S2. Feature Description

The dataset contains two epidemiological features, both measured as weekly case counts aggregated at the national level. Each feature corresponds to a specific population compartment in the TV-SIDARHE epidemic model used in this study. The table below provides a full description of each feature, its unit, and its epidemiological meaning.

Feature	Column Name	Notation in Model	Unit	Description
Confirmed Positive Cases (PCR-tested)	`D`	D (Diagnosed)	Number of persons per week	Weekly count of newly confirmed COVID-19 cases detected through Polymerase Chain Reaction (PCR) testing, including primarily asymptomatic or pre-symptomatic individuals identified through routine or targeted testing. Corresponds to the Diagnosed (D) compartment in the TV-SIDARHE model: detected asymptomatic infected individuals. This is the higher-volume series and serves as the dominant monitoring signal in Phase II.
Reported Symptomatic Cases	`R`	R (Recognized)	Number of persons per week	Weekly count of confirmed COVID-19 cases with documented symptomatic clinical presentation reported to health authorities. Corresponds to the Recognized (R) compartment in the TV-SIDARHE model: detected symptomatic infected individuals. This variable is the primary target for model fitting and prediction performance evaluation.

Note on measurement units: Both D and R are reported as absolute weekly case counts (number of persons per week) and represent newly recorded cases in each epidemiological week, not cumulative totals. This differs from the China dataset, where the reported figures are cumulative daily counts. The weekly aggregation reflects the reporting cadence of the RKI surveillance system.

S3. Descriptive Statistics

S3.1 Full Dataset (78 Weeks, Phase II Period)

Descriptive statistics for the complete 78-week dataset are reported in Table S3. The wide range and high standard deviation of both D and R reflect the multi-wave epidemic trajectory observed in Germany over this period, which includes the major Omicron-driven peak in early 2022, multiple secondary resurgences, and the subsequent stabilisation phase through mid-2023.

Feature	Min	Max	Mean	Median	Std. Dev.
Confirmed Cases, D (persons/week)	3,328	1,381,815	354,069	246,462	344,072
Symptomatic Cases, R (persons/week)	7,474	1,589,070	431,353	289,846	402,346

S3.2 First-Order Differenced Series

As confirmed by stationarity testing (Table 7 of the main paper), the raw D and R series are *directly stationary* in the German dataset — in contrast to the China case where first-order differencing was required. Nevertheless, first-order differenced series ΔD and ΔR were computed and verified as additionally stationary under both ADF and KPSS tests. Descriptive statistics for the differenced series are provided below for completeness.

Feature	Min	Max	Mean	Std. Dev.
ΔD (week-on-week change in D)	−269,153	+308,348	−3,591	94,862
ΔR (week-on-week change in R)	−310,437	+318,349	−3,426	110,798

S3.3 Stationarity Test Results

Prior to Phase II SPC monitoring, the stationarity of D and R was assessed using the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test, both at the 5% significance level with lag lengths selected via AIC. Unlike the China case, both raw series in the German dataset are directly stationary: the ADF test rejects the unit root null hypothesis and the KPSS test fails to reject the stationarity null hypothesis for both D and R. This is attributable to the structure of the German dataset, which spans a complete epidemic wave from peak to stabilisation, imparting mean-reverting characteristics to the raw series. The stationarity test results (Table 7 of the main paper) are reproduced below for reference.

Variable	ADF Statistic	p-value	KPSS Statistic	p-value	Conclusion
D (raw)	−4.984	< 0.001	0.089	0.100	Directly stationary (no differencing required)

Variable	ADF Statistic	p-value	KPSS Statistic	p-value	Conclusion
R (raw)	−3.858	0.014	0.079	0.100	Directly stationary (no differencing required)
ΔD	−5.891	< 0.001	0.103	0.100	Strictly stationary
ΔR	−6.421	< 0.001	0.093	0.100	Strictly stationary

S4. Data Partitioning and Usage

The Germany dataset supports two distinct analytical modules within the ensemble framework: (i) TV-SIDARHE epidemic model fitting and prediction, and (ii) SPC control chart Phase I baseline construction and Phase II online monitoring. Each partition is described in detail below.

S4.1 SPC Control Chart Partitions

S4.1.1 Phase I Training Set (Not in Provided File)

The Phase I training set covers **Week 10 of 2020 to Week 44 of 2021** (approximately 88 weeks) and is used to construct the in-control (IC) reference distribution for all control charts (RTSVDD, Autoencoder, and LSTM). This period corresponds to the initial containment and first-wave stabilisation phase of the COVID-19 pandemic in Germany, prior to the large-scale Omicron resurgence. This segment is not included in the provided CSV file.

Attribute	Value
Period	Week 10 of 2020 – Week 44 of 2021 (approximately 88 weeks)
Data included in provided file	No (historical in-control period; available from SurvStat@RKI 2.0)
Role	Phase I in-control baseline for RTSVDD, Autoencoder, and LSTM control charts

S4.1.2 Phase II Monitoring Set (Provided File)

The Phase II monitoring set covers **Week 45 of 2021 to Week 18 of 2023** (78 weeks), constituting the entirety of the provided CSV file. This period is used for online monitoring by all three control charts (RTSVDD, Autoencoder, LSTM). Of these 78 weeks, the segment from Week 45/2021 to Week 51/2022 (55 weeks) consists of real observational data, while the segment from Week 52/2022 to Week 18/2023 (23 weeks) is replaced by the predictive outputs of the TV-SIDARHE model within the ensemble framework.

Attribute	Value
Period	Week 45 of 2021 – Week 18 of 2023
Total weeks	78
D: range (persons/week)	3,328 – 1,381,815
D: mean ± std	354,069 ± 344,072
R: range (persons/week)	7,474 – 1,589,070
R: mean ± std	431,353 ± 402,346
Role	Phase II online monitoring for all three ensemble control charts

S4.2 TV-SIDARHE Model Partitions

Within the provided 78-week file, the data are further partitioned for TV-SIDARHE model fitting and prediction evaluation.

S4.2.1 Model Training Set

The TV-SIDARHE training set spans **Week 45 of 2021 to Week 51 of 2022** (55 weeks). It covers the major Omicron outbreak peak (around Week 10/2022) and the subsequent multi-wave resurgence and partial containment through the end of 2022.

Attribute	Value
Period	Week 45 of 2021 – Week 51 of 2022
Number of observations	55 weeks
D: range (persons/week)	132,017 – 1,381,815
D: mean ± std	419,337 ± 313,545
R: range (persons/week)	169,945 – 1,589,070
R: mean ± std	504,282 ± 366,574
Role	TV-SIDARHE and SIDARTHE model fitting; parameter optimisation via least squares (10,000 iterations)

S4.2.2 Model Test / Prediction Set

The TV-SIDARHE test set covers **Week 52 of 2022 to Week 18 of 2023** (23 weeks). The real observations for this period exist in the provided dataset and serve as ground truth for evaluating prediction accuracy. Within the ensemble framework, the TV-SIDARHE model's forecasted D and R trajectories for this segment replace the real observations as input to the Phase II SPC monitoring module.

Attribute	Value
Period	Week 52 of 2022 – Week 18 of 2023
Number of observations	23 weeks
D: range (persons/week)	3,328 – 1,094,701
D: mean ± std	197,993 ± 370,034
R: range (persons/week)	7,474 – 1,310,822
R: mean ± std	256,959 ± 437,794
Role	Prediction evaluation (R^2, MAPE) using real data as ground truth; model-predicted values used as SPC input in the ensemble framework

S4.3 Phase II Monitoring Stage Breakdown

Following the structure described in Section 5.2.1 of the main paper, the 78-week Phase II monitoring period is divided into three epidemiological stages. Stage 1 encompasses the period leading up to and including the major Omicron peak (around Week 10/2022); Stage 2 covers the post-peak resurgence period through the end of the real-data segment (Week 51/2022); Stage 3 corresponds to the period supplied by the TV-SIDARHE model predictions (Week 52/2022 – Week 18/2023), during which the epidemic progressively approaches a steady state.

Stage	Period	Weeks	Data Type	D: range	R: range	Epidemic Context
Stage 1	Week 45/2021 – Week 10/2022	10	Real observational data	202,569 – 1,215,712	193,912 – 1,414,667	Major Omicron-driven outbreak; rapid escalation to peak. First sustained OC alarms issued by RTSVDD (as early as Week 46/2021).
Stage 2	Week 11/2022 – Week 51/2022	45	Real observational data	132,017 – 1,381,815	169,945 – 1,589,070	Post-peak multiple resurgences under evolving policy interventions and seasonal variability; gradual trend toward stabilisation by end of 2022.
Stage 3	Week 52/2022 – Week 18/2023	23	TV-SIDARHE model predictions (real data available as ground truth	3,328 – 1,094,701	7,474 – 1,310,822	Epidemic approaches steady state; predicted trajectories supplied by TV-SIDARHE model serve as SPC

Stage	Period	Weeks	Data Type	D: range	R: range	Epidemic Context
			only)			inputs; real observations used for prediction accuracy evaluation only.

Note: The stage boundaries above are derived from the paper's description (Section 5.2.1) and the monitoring chart in Figure 13. Stage 1 and Stage 2 are separated at approximately Week 10/2022 (the major infection peak), while Stage 2 and Stage 3 are separated at Week 52/2022 (the transition from real to model-predicted data). The D and R ranges reported for Stage 3 reflect the real observations available for that period; the actual values processed by the control charts during Stage 3 are model-generated.

S5. Preprocessing Steps

The following preprocessing steps were applied to the raw German data prior to model fitting and control chart monitoring. All steps are fully described to ensure reproducibility.

S5.1 Data Extraction and Aggregation

Weekly case counts for Confirmed Positive Cases (D) and Reported Symptomatic Cases (R) were extracted from the *SurvStat@RKI 2.0* online database using ISO calendar week identifiers. The data are reported at the national level (all German federal states combined) and require no further geographic aggregation. No imputation was performed as the dataset contains no missing values.

S5.2 Stationarity Assessment

Unlike the China dataset, which required first-order differencing to achieve stationarity, both raw D and R series in the German dataset passed direct stationarity testing. This property is consistent with the mean-reverting structure of the data, which spans a complete epidemic wave including both the rise to peak and the subsequent decline. The ADF test rejects the unit root null hypothesis ($p < 0.001$ for D; $p = 0.014$ for R), and the KPSS test fails to reject the stationarity null hypothesis (both $p = 0.100$). Raw D and R were therefore used directly as input to the SPC monitoring module without differencing.

S5.3 Initial Condition Specification for TV-SIDARHE and SIDARTHE

For TV-SIDARHE and SIDARTHE model fitting, the initial compartment sizes were set to reflect the epidemic state at the beginning of the training period (Week 45/2021). The total population was fixed at $\mathbf{N = 8{,}320 \times 10^4}$ (83.2 million). The Susceptible (S) compartment was initialised as the residual population $\mathbf{S = N - I - D - A - R - H - E}$ (minus T for SIDARTHE) to satisfy the compartment conservation identity. Model parameters were estimated via least squares optimisation over 10,000 iterations.

Compartment	Symbol	Initial Value
Infected (undetected asymptomatic)	I	200,000
Diagnosed (detected asymptomatic)	D	279,857 (= observed D at Week 45/2021)
Ailing (undetected symptomatic)	A	100,000
Recognized (detected symptomatic)	R	271,241 (= observed R at Week 45/2021)
Healed	H	300,000
Extinct	E	20,000
Threatened (SIDARTHE only)	T	1,000
Susceptible (derived)	S	N − all others
Total population	N	$8{,}320 \times 10^4$

S6. Consolidated Data Partition Summary

Table S6 consolidates all data partitions described in Sections S4 and S5, providing a single reference table for the complete experimental setup of the Germany case study.

Partition	Period	n (weeks)	Features Used	Purpose
Phase I (SPC)	Week 10/2020 – Week 44/2021	~88 (not in file)	D, R	IC baseline for RTSVDD, Autoencoder, LSTM control charts
Phase II (SPC monitoring)	Week 45/2021 – Week 18/2023	78	D, R	Online monitoring for all three ensemble control charts
TV-SIDARHE training	Week 45/2021 – Week 51/2022	55	D, R	TV-SIDARHE and SIDARTHE model fitting; parameter optimisation
TV-SIDARHE test	Week 52/2022 – Week 18/2023	23	D, R (real, as ground truth)	Prediction evaluation (R^2, MAPE); real data serve as

Partition	Period	n (weeks)	Features Used	Purpose
				held-out ground truth only
Phase II Stage 1 (peak)	Week 45/2021 – Week 10/2022	10	D, R (real)	Major Omicron outbreak monitoring; first OC alarms issued
Phase II Stage 2 (post-peak)	Week 11/2022 – Week 51/2022	45	D, R (real)	Multi-wave resurgence monitoring; adaptive control limit adjustment
Phase II Stage 3 (predicted)	Week 52/2022 – Week 18/2023	23	D, R (model output)	TV-SIDARHE forecast as SPC input; epidemic stabilisation monitoring

Reference: Chukhrova, N., Plate, O., & Johannssen, A. (2024). Monitoring epidemic processes under political measures. *Statistics in Medicine*, 43, 2122–2160.